

Personalised therapy and target report

# Screening report

Retinal screening to therapy options, target priorities and trials

|                   |                                                                         |
|-------------------|-------------------------------------------------------------------------|
| Patient           | 64 years                                                                |
| Current medicines | pioglitazone, semaglutide                                               |
| Model             | insightrx-dinov2L-lora-ens2-141f4d6893 (live)                           |
| Status            | Unsaved screening (held 1 hour). Save the patient to consult and track. |

## Summary

- Retina:** Referable DR signal.
- Findings:** Referable diabetic retinopathy, Macular edema signal, Blood pressure and vascular risk.
- Therapy options:** 6 guideline classes; **2** major alert(s) on current medicines.
- Top target:** VEGFA (priority 98): Referable diabetic retinopathy (validated retinal model); Anti-VEGF intravitreal therapy: specialist-administered; 6 approved drugs; drug-contact site pLDDT 95 (PDB 1CZ8).
- Trials:** 9 recruiting trial(s) matched by finding.
- CMS diabetic eye exam gap:** open. Screening captured; the gap closes once an eye care professional reads it.

## Retina

| Eye       | Result              | Referable DR score | Highest ICDR grade | Macular edema signal |
|-----------|---------------------|--------------------|--------------------|----------------------|
| Right eye | Referable DR signal | 1.00               | 2                  | yes                  |
| Left eye  | Referable DR signal | 1.00               | 2                  | yes                  |



Right eye, quality assessable, DR score 1.00



Left eye, quality assessable, DR score 1.00

## Whole-body signals

Research association models from the retina plus patient details; history recorded by the clinician takes precedence. Red bar: score at or above the frozen threshold (black tick).

| Condition                         | Output                | Score vs threshold     | Reliability                                                                           |
|-----------------------------------|-----------------------|------------------------|---------------------------------------------------------------------------------------|
| Hypertension                      | Research signal       | <div><div></div></div> | Passes the release gate: cross-validated AUROC 0.67 (95% CI 0.64 to 0.70)             |
| Diabetic kidney involvement       | No exploratory signal | <div><div></div></div> | Exploratory: cross-validated AUROC 0.60 (95% CI 0.51 to 0.68), below the release gate |
| Diabetes-related vascular disease | No exploratory signal | <div><div></div></div> | Exploratory: cross-validated AUROC 0.59 (95% CI 0.54 to 0.63), below the release gate |
| Prior myocardial infarction       | No exploratory signal | <div><div></div></div> | Exploratory: cross-validated AUROC 0.62 (95% CI 0.56 to 0.67), below the release gate |

| Condition                                                       | Output                | Score vs threshold | Reliability                                                                           |
|-----------------------------------------------------------------|-----------------------|--------------------|---------------------------------------------------------------------------------------|
| <b>Diabetic neuropathy</b>                                      | No exploratory signal |                    | Exploratory: cross-validated AUROC 0.59 (95% CI 0.52 to 0.66), below the release gate |
| <b>Diabetic foot</b>                                            | No exploratory signal |                    | Exploratory: cross-validated AUROC 0.61 (95% CI 0.57 to 0.66), below the release gate |
| <b>Obesity</b>                                                  | Near-chance score     |                    | Near chance: cross-validated AUROC 0.53 (95% CI 0.47 to 0.58 includes 0.5)            |
| <b>Cardiovascular composite (vascular disease or prior MI)</b>  | No exploratory signal |                    | Exploratory: cross-validated AUROC 0.59 (95% CI 0.55 to 0.63), below the release gate |
| <b>Diabetes complications composite (kidney, nerve or foot)</b> | No exploratory signal |                    | Exploratory: cross-validated AUROC 0.55 (95% CI 0.51 to 0.59), below the release gate |

## Therapy options to consider

Guideline classes for this patient's findings, ranked by guideline strength only (sponsorship never enters the ranking). Current medicines: pioglitazone, semaglutide.

| Therapy class                                     | Why (findings)                                       | Evidence tier           | Example drugs                                          | Already on  | Source                                                                                     |
|---------------------------------------------------|------------------------------------------------------|-------------------------|--------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------|
| <b>Tighten glucose, blood pressure and lipids</b> | Referable diabetic retinopathy                       | Guideline first-line    | -                                                      | -           | ADA Standards of Care in Diabetes, Section 12 (Retinopathy)                                |
| <b>Anti-VEGF intravitreal therapy</b>             | Referable diabetic retinopathy, Macular edema signal | Specialist-administered | aflibercept, ranibizumab, faricimab, bevacizumab       | -           | AAO Diabetic Retinopathy Preferred Practice Pattern; DRCR Retina Network Protocols T and S |
| <b>Fenofibrate (adjunct)</b>                      | Referable diabetic retinopathy                       | Trial evidence          | fenofibrate                                            | -           | FIELD (Lancet 2007); ACCORD-Eye (NEJM 2010); LENS (NEJM Evidence 2024)                     |
| <b>ACE inhibitor or ARB</b>                       | Blood pressure and vascular risk                     | Guideline first-line    | lisinopril, ramipril, losartan, valsartan, candesartan | -           | ADA Standards of Care, Sections 10 and 11; KDIGO 2022 Diabetes in CKD                      |
| <b>SGLT2 inhibitor</b>                            | Blood pressure and vascular risk                     | Guideline first-line    | empagliflozin, dapagliflozin, canagliflozin            | -           | KDIGO 2022 Diabetes in CKD; ADA Standards of Care, Section 11                              |
| <b>GLP-1 receptor agonist</b>                     | Blood pressure and vascular risk, Diabetes control   | Guideline first-line    | semaglutide, dulaglutide, liraglutide                  | semaglutide | ADA Standards of Care, Sections 9 and 10; SUSTAIN-6; FLOW (NEJM 2024)                      |

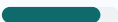
## Interaction and drug-disease alerts

| Severity     | Alert                              | Involves                                                     | Detail and source                                                                                                                                                                                                                                                                        |
|--------------|------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Major</b> | Diabetic retinopathy complications | semaglutide + dulaglutide + liraglutide (already prescribed) | Semaglutide was associated with more DR complications in SUSTAIN-6, linked to rapid glucose lowering. With a retinopathy signal, arrange retinal follow-up before and after starting. Semaglutide label, Warnings: diabetic retinopathy complications; ADA Standards of Care, Section 12 |
| <b>Major</b> | Macular edema with pioglitazone    | pioglitazone (already prescribed)                            | Thiazolidinediones have been associated with new or worsening macular edema. With a macular edema signal, review the need for pioglitazone and refer. Pioglitazone label, Warnings: macular edema                                                                                        |

## Personalised target priorities

Where this patient's phenotype points at the molecular level. Priority = 100 x phenotype link x (0.5 guideline actionability + 0.3 structural tractability at the drug site + 0.2 clinical fit). Actionability comes from US guideline strength, so drugs approved only abroad do not lift a target. It ranks where to look; it is not a treatment recommendation.

**Top target for this patient: VEGFA** (Vascular endothelial growth factor A), priority 98. Referable diabetic retinopathy (validated retinal model); Anti-VEGF intravitreal therapy: specialist-administered; 6 approved drugs; drug-contact site pLDDT 95 (PDB 1CZ8).

| Target                                                           | Priority                                                                               | Linked findings                                      | Drugs                     | Structure                                                           | Rationale                                               |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------|---------------------------------------------------------------------|---------------------------------------------------------|
| <b>VEGFA</b><br>Vascular endothelial growth factor A             |  98   | Referable diabetic retinopathy; Macular edema signal | 6 approved<br>3 pipeline  | drug-contact site pLDDT 95 (PDB 1CZ8)                               | Classes: Anti-VEGF intravitreal therapy                 |
| <b>AGTR1</b><br>Angiotensin II receptor type 1                   |  84   | Blood pressure and vascular risk                     | 7 approved<br>0 pipeline  | drug-contact site pLDDT 95 (PDB 4ZUD)                               | Classes: ACE inhibitor or ARB                           |
| <b>ACE</b><br>Angiotensin-converting enzyme                      |  83   | Blood pressure and vascular risk                     | 10 approved<br>0 pipeline | drug-contact site pLDDT 93 (PDB 1O86)                               | Classes: ACE inhibitor or ARB                           |
| <b>SLC5A2</b><br>Sodium/glucose cotransporter 2 (SGLT2)          |  83   | Blood pressure and vascular risk                     | 8 approved<br>0 pipeline  | drug-contact site pLDDT 92 (PDB 7VSI)                               | Classes: SGLT2 inhibitor                                |
| <b>ANGPT2</b><br>Angiopoietin-2                                  |  80   | Macular edema signal                                 | 1 approved<br>7 pipeline  | 81% of residues confidently predicted (no drug complex in this set) | Classes: Anti-VEGF intravitreal therapy                 |
| <b>PPARA</b><br>Peroxisome proliferator-activated receptor alpha |  77   | Referable diabetic retinopathy                       | 8 approved<br>2 pipeline  | 75% of residues confidently predicted (no drug complex in this set) | Classes: Fenofibrate (adjunct)                          |
| <b>AKR1B1</b><br>Aldose reductase                                |  74  | Referable diabetic retinopathy                       | 2 approved<br>2 pipeline  | drug-contact site pLDDT 98 (PDB 4JIR)                               | No guideline class yet                                  |
| <b>GLP1R</b><br>Glucagon-like peptide-1 receptor                 |  73 | Diabetes control; Blood pressure and vascular risk   | 7 approved<br>3 pipeline  | drug-contact site pLDDT 86 (PDB 7KIO)                               | Engaged by semaglutide. Classes: GLP-1 receptor agonist |

## Recruiting clinical trials

ClinicalTrials.gov, searched by finding near the clinic; no patient data is sent. The site confirms full eligibility.

| Finding                        | Trial                                                                                                                                                                                                                                                     | Phase / sponsor                          | Pre-check                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|
| Referable diabetic retinopathy | <b>A Study to Investigate the Safety, Tolerability, and Pharmacokinetics of RO7663498 Following Intravitreal Administration in Participants With Diabetic Retinopathy</b><br><a href="#">NCT07588100</a> Global Research Management, Glendale, California | Phase 1 · Hoffmann-La Roche              | Age and sex fit; confirm full eligibility |
| Referable diabetic retinopathy | <b>Implementing AI-based Glaucoma Screening Within Federally Qualified Health Centers</b><br><a href="#">NCT06629649</a> Department of Ophthalmology and Visual Sciences, University of Alabama at Birmingham, Birmingham, Alabama                        | NA · University of Alabama at Birmingham | Age and sex fit; confirm full eligibility |
| Referable diabetic retinopathy | <b>Longitudinal Natural History Study of Retinal Function in Eyes of Patients With Diabetes</b><br><a href="#">NCT07270133</a> Southeast Retina Center, P.C., Augusta, Georgia                                                                            | N/A · Jaeb Center for Health Research    | Age and sex fit; confirm full eligibility |
| Macular edema signal           | <b>A Study in People With Diabetic Macular Edema to Test How Well Different Doses of BI 3812465 Are Tolerated</b><br><a href="#">NCT07553429</a> Associated Retina Consultants, Ltd., Phoenix, Arizona                                                    | Phase 1, Phase 2 · Boehringer Ingelheim  | Age and sex fit; confirm full eligibility |

| Finding                          | Trial                                                                                                                                                                                                                                                                | Phase / sponsor                          | Pre-check                                 |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|
| Macular edema signal             | <b>A Study to Evaluate the Safety, Tolerability, Pharmacodynamics, and Pharmacokinetics of RO7823653 in Participants With Diabetic Macular Edema (DME)</b><br><a href="#">NCT07425522</a> Associated Retina Consultants - Peoria - DocTrials - PPDS, Peoria, Arizona | Phase 1 · Genentech, Inc.                | Age and sex fit; confirm full eligibility |
| Macular edema signal             | <b>A Clinical Trial of MK-8748 Compared to Aflibercept in Participants With Diabetic Macular Edema (MK-8748-004)</b><br><a href="#">NCT07705555</a> Global Research Management ( Site 0066), Glendale, California                                                    | Phase 3 · Merck Sharp & Dohme LLC        | Age and sex fit; confirm full eligibility |
| Blood pressure and vascular risk | <b>Implementing AI-based Glaucoma Screening Within Federally Qualified Health Centers</b><br><a href="#">NCT06629649</a> Department of Ophthalmology and Visual Sciences, University of Alabama at Birmingham, Birmingham, Alabama                                   | NA · University of Alabama at Birmingham | Age and sex fit; confirm full eligibility |
| Blood pressure and vascular risk | <b>Improving the Collaborative Health of Adult Dyads</b><br><a href="#">NCT05370014</a> University of South Carolina, Columbia, South Carolina                                                                                                                       | NA · University of South Carolina        | Age and sex fit; confirm full eligibility |
| Blood pressure and vascular risk | <b>EASi-PROTKT™ - A Study to Test Vicadrostat (BI 690517) Taken Together With Empagliflozin in People With Type 2 Diabetes, High Blood Pressure, and Cardiovascular Disease</b><br><a href="#">NCT07064473</a> Pinnacle Research Group, LLC, Anniston, Alabama       | Phase 3 · Boehringer Ingelheim           | Age and sex fit; confirm full eligibility |

## CMS quality and billing

| Measure / code                    | What it is                                                         | Fit for this encounter                                                                                                                                       |
|-----------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CMS131 / MIPS Quality #117</b> | Diabetes: Eye Exam                                                 | A screening read and signed by an eye care professional counts toward closing the gap. A signed specialist response in Insight Rx is the evidence to record. |
| <b>HEDIS EED</b>                  | Eye Exam for Patients With Diabetes                                | Closing EED gaps moves Star Ratings, which drive Medicare Advantage bonus payments to health plans.                                                          |
| <b>CPT 92227</b>                  | Retinal imaging, remote clinical staff review and report           | Store-and-forward capture with review by clinical staff (no physician interpretation).                                                                       |
| <b>CPT 92228</b>                  | Retinal imaging, remote physician or QHP interpretation and report | Insight Rx's workflow: photos captured at point of care, interpreted and signed by a physician.                                                              |
| <b>CPT 92229</b>                  | Retinal imaging, point-of-care automated analysis and report       | Autonomous AI with an FDA-cleared device. Insight Rx is not FDA-cleared, so this does not apply today.                                                       |

## Methods, sources and limits

- Retinal models: DINOv2-L multi-task ensemble trained on mBRSET (portable camera, dilated); other cameras are a domain shift.
- Therapy classes and interactions are curated from ADA, AAO, KDIGO guidelines and FDA labels, cited per row; not a complete interaction checker.
- Targets: UniProt; structures: AlphaFold DB (CC-BY 4.0) and RCSB PDB; drugs: ChEMBL mechanisms; trials: ClinicalTrials.gov.
- Prepared for Dr. Alex Morgan. Patient identity in this workspace is synthetic.